Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 9: Approved Solutions

Conjunctive Normal Form, Satisfiability, and SAT

Covers *MCS* Sections 3.3.2, 3.4.1, and 3.5

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Everything flips

The two recipes side by side

	Built from	Shape	Variable that is T in that row
DNF	**true** rows	OR of ANDs	appears plain
CNF	**false** rows	AND of ORs	appears *negated*

Count the F's before writing a CNF. That is how many clauses you will have.

Three words, and how they relate to Monday

Valid = true in every row = Chartrand's *tautology*.
Satisfiable = true in at least one row.
Unsatisfiable = true in no row = Chartrand's *contradiction*.
Every valid formula is satisfiable; the converse fails. And S is satisfiable if and only if $\sim S$ is not valid.

Three mistakes to expect

1. **Using the DNF sign convention.** In a CNF clause, a variable that is T in the row appears *negated*.
2. **Confusing valid with satisfiable.** "There is an assignment making it true" is a far weaker claim than "every assignment does."
3. **Claiming unsatisfiable after testing a few assignments.** That is the sampling error from Week 1, wearing a new hat.

Part II: Practice Problems

Section A — Reading CNF off a table

1. Build the truth table for $P \oplus Q$, then write its CNF.

P	Q	$P \oplus Q$	Clause
T	T	F	$(\sim P) \vee (\sim Q)$
T	F	T	—
F	T	T	—
F	F	F	$P \vee Q$

Approved solution

Two false rows, so two clauses:

$$P \oplus Q \;\equiv\; \big((\sim P) \vee (\sim Q)\big) \wedge \big(P \vee Q\big)$$

Read it aloud: *not both, and at least one* — which is precisely "P or Q but not both."
Compare Practice Set 2, Problem 12, where you verified $P \oplus Q \equiv (P \vee Q) \wedge \sim(P \wedge Q)$ by truth table. The CNF recipe produces the same statement with De Morgan already applied to the second factor.

2. Write the CNF of each.

(a) $P \wedge Q$

(b) $P \vee Q$

(c) $\sim P$ (over the variables P and Q)

Approved solution

(a) False in three rows, so three clauses: $\big((\sim P) \vee Q\big) \wedge \big(P \vee (\sim Q)\big) \wedge \big(P \vee Q\big)$.
(b) False in one row only (FF), so one clause: $P \vee Q$. The formula was already a CNF.
(c) False in two rows (TT and TF), so two clauses: $\big((\sim P) \vee (\sim Q)\big) \wedge \big((\sim P) \vee Q\big)$.

Compare (a) with Practice Set 8, Problem 2(a), where the *DNF* of $P \wedge Q$ was a single term. Same formula: one term in DNF, three clauses in CNF. Which form is shorter depends entirely on whether there are fewer T's or fewer F's.

Section B — Validity and satisfiability

3. Classify each as valid, satisfiable but not valid, or unsatisfiable. Justify briefly.

(a) $(P \Rightarrow Q) \vee (Q \Rightarrow P)$

(b) $P \wedge (\sim P)$

(c) $(P \vee Q) \wedge (\sim P)$

(d) $\big(P \wedge (P \Rightarrow Q)\big) \Rightarrow Q$

Approved solution

(a) Valid. True in all four rows — you classified this in Practice Set 7.
(b) Unsatisfiable. A contradiction; no row makes it true.
(c) Satisfiable but not valid. True at P false, Q true; false at, for instance, P true, Q true.
(d) Valid. This is modus ponens, which you verified Monday.

Part (c) is the one that needs care. Showing it satisfiable takes one assignment; showing it not valid takes one more. Two different witnesses, one for each half of the claim — and neither is a general argument.

4. Determine whether each CNF formula is satisfiable. Give a satisfying assignment or argue that none exists.

(a) $(P \vee Q) \wedge (\sim P \vee Q) \wedge (\sim Q)$

(b) $(P \vee Q) \wedge (\sim P \vee R) \wedge (\sim Q \vee \sim R)$

Approved solution

(a) Unsatisfiable. The third clause forces Q false. With Q false, the first clause forces P true and the second forces P false. Contradiction, so no assignment works.
Note this argument checked no rows at all — it propagated forced values. That is roughly how real SAT solvers work.
(b) Satisfiable. Take P true, Q false, R true. Then $(P \vee Q)$ is true, $(\sim P \vee R)$ is $F \vee T$, true, and $(\sim Q \vee \sim R)$ is $T \vee F$, true. Other assignments work too.

The asymmetry again. Part (b) needed one assignment. Part (a) needed an argument ruling out all eight. Satisfiable is the existential side and unsatisfiable is the universal side — exactly the pattern from Week 1.

Section C — Specifications

5. A build system must satisfy all three requirements at once:

(1) If the tests pass, the artifact is published.

(2) The artifact is not published, or the version is tagged.

(3) The tests pass, and the version is not tagged.

Let T, A, and V stand for the three claims. Write each requirement as a clause, then determine whether all three can hold together.

Approved solution

(1) $T \Rightarrow A \equiv (\sim T) \vee A$ by Theorem 1.48.
(2) $(\sim A) \vee V$.
(3) $T \wedge (\sim V)$ — this is two unit clauses, T and $\sim V$.
The conjunction is $\big((\sim T) \vee A\big) \wedge \big((\sim A) \vee V\big) \wedge T \wedge (\sim V)$.
Not satisfiable. Requirement (3) forces T true and V false. With T true, clause (1) forces A true. With A true, clause (2) forces V true — contradicting V false. No assignment satisfies all three.

This is what SAT is for in practice. A set of specifications is implementable only if their conjunction is satisfiable. Here the specification is self-contradictory, and no amount of clever engineering will produce a system meeting it. Discovering that *before* writing code is the whole value.

Section D — Writing

6. In five to eight complete sentences, explain why building a truth table is an impractical way to solve SAT for large formulas, and why this matters beyond this course. Use a specific number of variables in your explanation.

Approved solution

Model answer. A formula with n variables has a truth table with 2^n rows, because each variable independently takes one of two values. That growth is manageable for small n but not for long: at $n = 30$ the table already has more than a billion rows, and at $n = 300$ it has more rows than there are atoms in the observable universe. So checking every assignment is not a usable method once formulas reach realistic size, even though it is perfectly correct. Nobody has found a fundamentally faster general method, and nobody has proved that none exists — that open question is P versus NP, and SAT was the first problem shown to be NP-complete. This matters beyond the course because SAT solvers are used in practice for hardware verification, scheduling, and program analysis, where they handle formulas with millions of variables by exploiting structure rather than enumerating rows.

Full credit requires a specific n with its row count, and the point that correctness and practicality are different things. Mentioning P versus NP is a bonus, not a requirement.

Section E — Challenge (optional)

7. Show how to obtain a CNF for a formula from the DNF of its *negation*.

Approved solution

Let S be the formula. Build the DNF of $\sim S$: one AND-term for each row where $\sim S$ is true — that is, each row where S is **false**.
Now negate both sides. $S \equiv \sim(\sim S)$ by double negation, so S is equivalent to the negation of that DNF. Applying De Morgan turns the outer OR into an AND and each inner AND-term into an OR-clause, with every literal flipped:

$$\sim\Big(\bigvee_i (\ell_{i1} \wedge \cdots \wedge \ell_{in})\Big) \equiv \bigwedge_i \big((\sim \ell_{i1}) \vee \cdots \vee (\sim \ell_{in})\big).$$

The result is exactly the CNF of S: one clause per false row of S, with each variable negated if it was T in that row.

This explains the sign flip that the recipe just asserts. The CNF rule looks backwards only because De Morgan has been applied silently. It is the same law you proved in Week 1, now organising an entire construction.

Instructor notes

- **Problem 6 is the set's centrepiece.** Three plausible build requirements that cannot all hold. It shows what SAT is *for* better than any abstract statement, and it is worth ten minutes at the board.
- **Problem 5(a)** is solved by propagating forced values, not by enumeration. Point that out — it is a preview of how real solvers work and it rewards the students who did not build a table.
- **Problem 2 paired with Practice Set 8 Problem 2** shows the same formulas in both normal forms. $P \wedge Q$: one DNF term, three CNF clauses. That contrast is the cleanest way to make "fewer T's or fewer F's" concrete.
- **Problem 8** explains the sign flip that the CNF recipe otherwise just asserts. Strong students find the recipe arbitrary until they see this.
- **Problem 4(c)** needs two separate witnesses. Watch for students who give one assignment and think they have answered both halves.
- No textbook exercises exist for this material — Chartrand omits normal forms. Point students to *MCS* Sections 3.4–3.5 on Canvas.